

EXPERIMENT 2

Draw the Incident Response Life Cycle Diagram and Describe Each Phase with a Real-Life Example (WannaCry Ransomware)

Aim

To study the Incident Response Life Cycle and understand how an organization prepares for, detects, contains, eradicates, recovers from, and learns from cybersecurity incidents using the WannaCry ransomware attack as a real-world example.

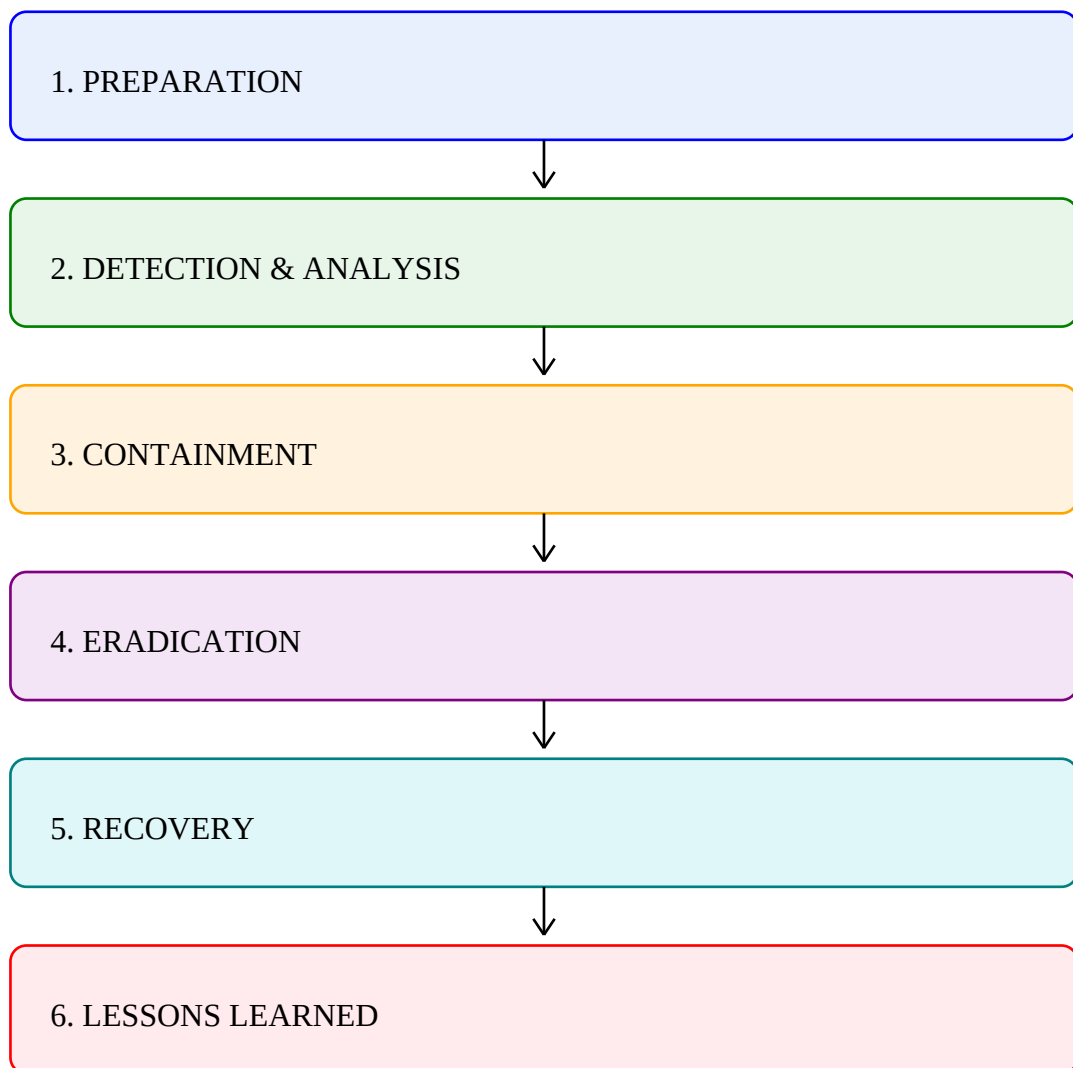
Objectives

- Understand the six phases of the Incident Response Life Cycle.
- Study the WannaCry ransomware incident.
- Learn the activities performed during each response phase.
- Understand how proper incident response minimizes business impact.

Introduction

Incident Response (IR) is a systematic process followed by organizations to identify, analyze, respond to, manage, and recover from cybersecurity incidents. The primary objective is to minimize financial loss, reduce operational downtime, preserve digital evidence, protect confidential information, and restore normal business operations as quickly as possible. Modern organizations follow frameworks such as the NIST Incident Response Life Cycle, which consists of six phases: Preparation, Detection & Analysis, Containment, Eradication, Recovery, and Lessons Learned.

Incident Response Life Cycle Diagram



Real-Life Example: WannaCry Ransomware (2017)

WannaCry was a global ransomware attack that began in May 2017. It exploited the EternalBlue vulnerability in the Windows SMB protocol and infected more than 230,000 computers across over 150 countries. Victims included hospitals, businesses, government agencies, and manufacturing companies. Files were encrypted and victims were asked to pay a Bitcoin ransom. Organizations with updated systems and reliable backups recovered much faster than those without them.

1. Preparation

Definition: Preparation is the process of establishing policies, procedures, security controls, backup strategies, and trained personnel before an incident occurs. Organizations deploy antivirus

software, configure firewalls, apply security patches, maintain offline backups, and educate employees about phishing attacks. This phase reduces the likelihood and impact of cyberattacks.

Real-Life Example: Microsoft released the MS17-010 patch before the WannaCry outbreak. Organizations that installed it and maintained backups were largely protected.

2. Detection & Analysis

Definition: This phase focuses on identifying suspicious activity, confirming whether a security incident has occurred, determining the attack vector, identifying affected systems, and assessing the severity of the incident using security logs, SIEM, EDR, and forensic analysis.

Real-Life Example: Security teams observed encrypted files, ransomware notes, and abnormal SMB traffic, confirming a WannaCry infection.

3. Containment

Definition: Containment prevents the attack from spreading while preserving evidence. Short-term containment isolates infected systems immediately, while long-term containment allows essential business operations to continue securely.

Real-Life Example: Administrators disconnected infected computers, disabled SMBv1, and isolated affected networks to stop WannaCry from spreading.

4. Eradication

Definition: Eradication completely removes malware, closes exploited vulnerabilities, deletes malicious files, patches systems, and verifies that attackers no longer have access.

Real-Life Example: IT teams removed ransomware, applied the MS17-010 patch, and rebuilt compromised systems.

5. Recovery

Definition: Recovery restores systems and services to normal operation. Clean backups are restored, systems are monitored for reinfection, and users gradually resume business activities after validation.

Real-Life Example: Encrypted files were restored from offline backups and systems were safely reconnected.

6. Lessons Learned

Definition: After recovery, organizations document the incident, identify root causes, evaluate the effectiveness of the response, and improve future security controls, policies, and employee awareness.

Real-Life Example: Following WannaCry, many organizations improved patch management, network segmentation, backup policies, and cybersecurity training.

Observation

Phase	Observation
Preparation	Security patches and backups reduced risk.
Detection	Ransomware alerts and encrypted files identified.

Containment	Infected devices were isolated.
Eradication	Malware removed and vulnerabilities patched.
Recovery	Systems restored from backups.
Lessons Learned	Security controls strengthened.

Result

The Incident Response Life Cycle successfully demonstrates how organizations can prepare for, detect, contain, eradicate, recover from, and learn from cybersecurity incidents. The WannaCry attack emphasizes the importance of timely patch management, offline backups, continuous monitoring, and employee awareness.